Study of the multicritical point in the vortex phase diagram $(H\|c)$ of a weakly pinned crystal of $YBa_2Cu_3O_{7-\delta}$

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dc magnetization studies on a high quality $YBa_2Cu_3O_{7-\delta}$ crystal show that the flux-line lattice melting line extends beyond the notional ''tricritical point'' in the intermediate field region (~ 10 T), which broadly agrees with a recent theoretical proposal. The magnetization hysteresis data reveal the presence of both the peak effect (PE) and the fishtail effect in isothermal scans in the temperature interval 66–75 K. The broad anomaly in the magnetization hysteresis observed at lower temperatures ($T<50$ K), splits into two well separated peaks above 65 K. The upper peak closely resembles the classical PE phenomenon and the locus of the peak fields shows a turnaround behavior. On the other hand, the lower peak progressively moves to smaller fields as $T\rightarrow T_c$.

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Ever since the pioneering magnetization measurements in $YBa_2Cu_3O_{7-\delta}$ (YBCO) by Müller, Takashige, and Bednorz,[1] YBCO has become one of the most investigated systems with a view to understand the different vortex phases and transformations among them. The first significant theoretical framework for the transformation of the pristine Abrikosov vortex state was put forth by Nelson,[2] when he established the first-order melting of the flux-line lattice (FLL) into an entangled vortex liquid. These studies[1,2] led to a continuous development of the newer ideas on the vortex phase diagram of high-T_c superconductors. A contemporary view based on the experimental data and theoretical studies[3–6] on the vortex state suggests that the magnetic phase diagram of YBCO comprises phases, like, an ordered vortex solid (called a Bragg glass with no dislocations), a disordered solid (a vortex glass with proliferation of dislocations), a vortex liquid, etc. Further, the phase boundary, such as, the melting curve meets another phase boundary separating the ordered and disordered solids at the so called *tricritical point*, and above this point one ought to observe a second order transition to a vortex glass state. Such a phase diagram is expected on the basis of the competition among the elastic energy, the pinning energy and the thermal energy. The transition lines are determined by matching any of the two energies, and the match of all the three energies presumably locates the tricritical point.

However, recent studies by Nishizaki *et al.*[7] and Giller *et al.*[8] on high quality twinned and untwinned crystals of YBCO have shown that the vortex phase diagram is more complicated than that envisaged by current theories.[3–5] They have focussed on the anomalous second magnetization peak (usually called the fishtail peak or simply the FE) in the isothermal hysteresis loops near the *multicritical point* for an untwinned YBCO. Below this peak, the pinning force showed a steep increase at a characteristic field H^*, such that the locus of $H^*(T)$ is connected both with the first-order vortex melting line and the second-order vortex glass (topological) transition line at the *multicritical point*. A field driven transition from an ordered Bragg glass to the disordered vortex glass was ascribed as a possible origin of the onset of anomalous behavior at H^*. However, the nature of the transition [whether it is a phase transition or a crossover at $H^*(T)$] remains unclear. It is now widely accepted that the FLL melting in YBCO occurs at the upper edge of the peak effect (PE) phenomenon, which is a sharp increase in the critical current density J_c as function of H or T near the irreversibility line.[9–12] On the other hand, a broad peak occurs at lower temperatures [i.e., $t(=T/T_c)<0.9$], where thermal fluctuations effects are presumably weaker, and it is designated as a fishtail peak.[13] Kokkaliaris *et al.*[14] have arrived at the same conclusion by studying the memory effects (thermomagnetic history effects) in detwinned YBCO crystals using a procedure of the partial magnetization loops, although the question of the exact location of the *multicritical point* remained unresolved.

At this juncture, it is worth while to recall experimental results[15] and a recent theoretical work,[16] which have shown that the end point of the first order FLL melting appears to be well separated from (in fact extends beyond) the point where the topological transition allegedly meets the melting line. They have also shown[15] that the above mentioned behavior was present in both twinned and detwinned YBCO crystals. In view of these developments, we have undertaken to study the vortex phase diagram at intermediate fields in a crystal of YBCO,[17,18] with very low density of twins, in which we had demonstrated the occurrence of a reentrant characteristic in the locus of the peak temperatures of the PE at very low fields ($H<0.1$ T).[19] In this work, we report on the search and the identification of the anomalous variations in $J_c(H)$ and sudden changes in equilibrium magnetization values via the dc magnetization measurements. In the YBCO system, the presence of some twin boundaries is considered[7,9,12] to facilitate the occurrence (and detection) of the PE phenomenon. The isothermal magnetization hysteresis loops and

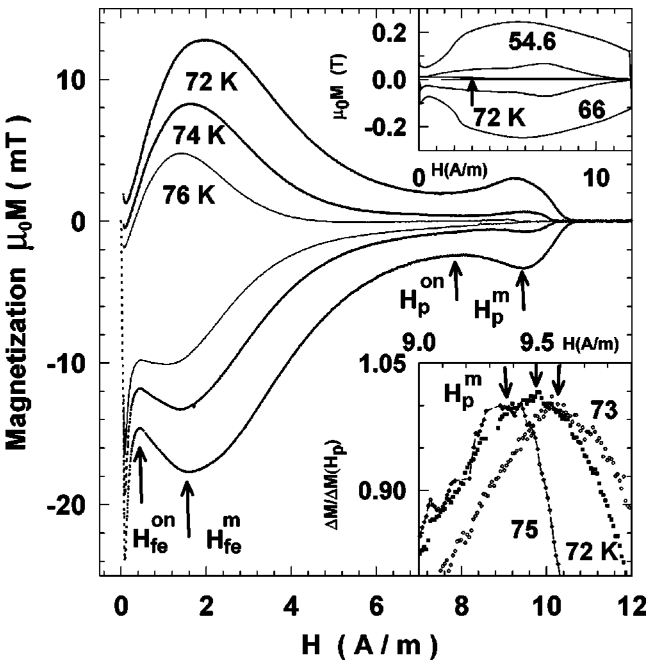

FIG. 1. Isothermal magnetization data in a single crystal of $YBa_2Cu_3O_7$ ($H \| c$) at 72, 74, and 76 K in the field range up to 12 T. The upper inset presents the M-H loops at lower temperatures, which essentially display the fishtail effect (FE). The lower inset shows the evolution of the reentrant characteristic in the maximum (H_p^m) of the classical peak effect (PE). Note the position of H_p^m at 72 K in the manner of those at 73 and 75 K.

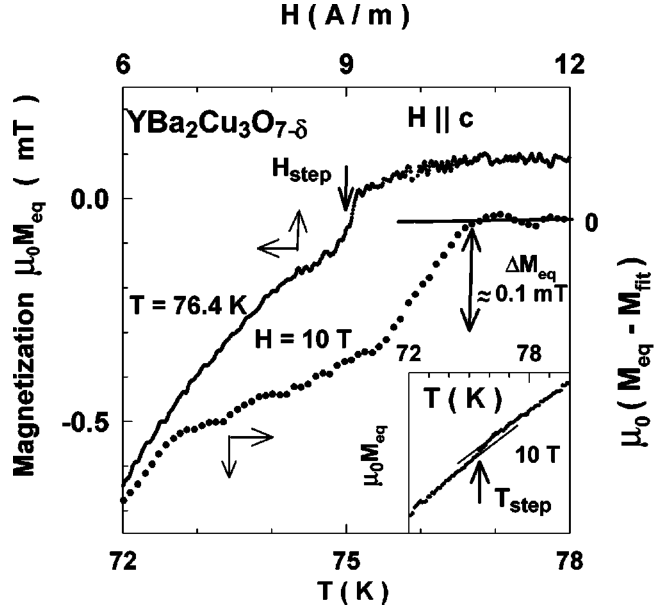

FIG. 2. The occurrence of a step change in the equilibrium magnetization (ΔM_{eq}) in an isothermal ($T = 76.4$ K) and an isofield ($H = 10$ T) scan are displayed together. The inset displays the M_{eq} vs T at 10 T, in which the position of the step change in M_{eq} near 75.5 K is identified via making two linear fits, one for $T < 75$ K and other for $T > 76$ K. The main panel shows the plot of $(M_{eq} - M_{fit})$ vs T, where the M_{fit} represents the fit to the data above 76 K.

isofield temperature dependent scans were performed using a vibrating sample magnetometer (Oxford Instruments, UK). Data were taken in both the field-cooled (FC) and the zero field-cooled (ZFC) modes.

The main panel of Fig. 1 shows the isothermal magnetization data at 72, 74, and 76 K, which unambiguously demonstrate the occurrence of the FE and the PE in the same isothermal scan. The two effects manifest as two maxima, the lower field one being the FE and the upper field one being the PE. In the narrow interval from 68 to 75 K, the peak field of the PE moves to higher values and the width as well as the height of the PE bubble diminish. This higher field anomaly is indistinguishable below 66 K, as it coalesces into the FE peak. The upper inset in Fig. 1 shows how the PE and the FE coalesce into a composite broad peak at low temperatures ($T < 65$ K). It is important to point out that the peak position of the FE does not vary appreciably in the temperature interval 50 to 65 K. On the other hand, the lower inset in Fig. 1 brings out the reentrant characteristic in the locus of the maximum (H_p^m) of the PE in the temperature region 70 to 75 K. Note that the H_p^m at 72 K lies in between the corresponding values at 73 and 75 K. The presence of such a reentrant behavior in $H_p^m(T)$ confirms that our sample is of high quality with very low-pinning force, as suggested by Nishizaki *et al.*[7]

Once the PE feature disappears above 76 K, a large reversible region surfaces above the second peak. We have made a successful search of the step change in the reversible (i.e., equilibrium) magnetization (ΔM_{eq}) in isothermal and isofield scans in the temperature interval 70 to 80 K and in fields upto 12 T, respectively as also reported by Nishizaki *et al.*[7] Figure 2 provides examples of ΔM_{eq} at $H_{step} \approx 9$ T in an isothermal scan at 76.4 K and at $T_{step} \approx 75.5$ K in an isofield scan at 10 T. The presence of $\Delta M_{eq}(T)$ reaffirms the notion[9] that the FLL melting transition occurs just above the PE anomaly. Our values of ΔM_{eq} are typically of the order of 0.1 mT corresponding to an entropy change per vortex (per CuO_2 double layer) of the order of $k_B T$, which compares favorably with the values of the entropy change in YBCO reported in the literature.[7,10]

Figure 3 portrays the ratio of the normalized values of the magnetization ($\propto J_c$) as a function of field at different temperatures ($54 \text{ K} \leqslant T \leqslant 72$ K). The inset of Fig. 3 shows the similar data on an expanded scale near the irreversibility line at $T \geqslant 72$ K. The data in the main panel show how the two anomalous maxima evolve into a single complex maximum as one progressively lowers the temperature. The inset panel exhibits the movement of the FE peak to lower fields at $T > 72$ K. The log-log plot of Fig. 3 can be utilized to mark out the field regions in which the power law decay in $J_c(H)$ holds at fields below the onset of the FE peak. Such a region could be identified with the collectively pinned ordered vortex lattice (akin to a Bragg glass phase[3]). Further, it may be pointed out that at $T > 66$ K, another power law behavior can be marked between the peak position of the FE (H_{fe}^m) and the onset of the PE (H_p^{on}). This second power law feature may indicate the existence of a weaker pinning region in the (H,T) phase diagram, which was also considered by Nishizaki *et al.*[7]

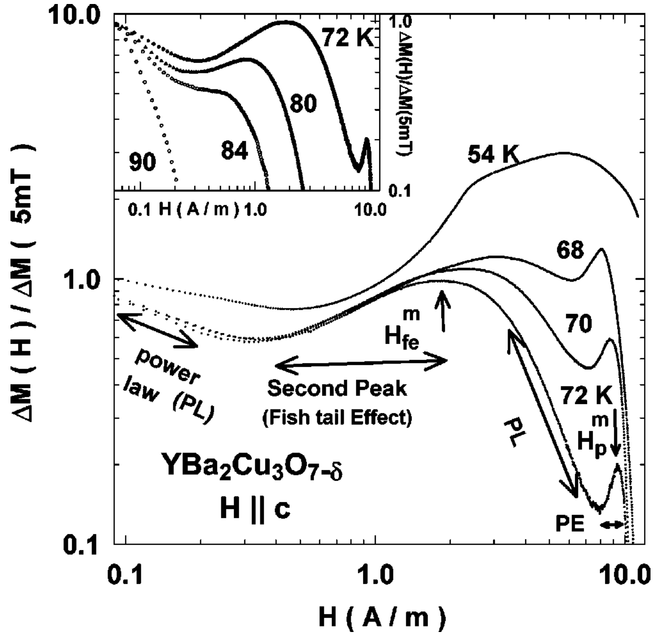

FIG. 3. Plot of the normalized J_c $[\alpha \Delta M(H)/\Delta M(5 \text{ mT})]$ versus H at various temperatures. The demarcation of the entire field scan at 72 K into different regions, such as power law, second peak (or FE), PE, etc., is shown along with the positions of the maxima of FE and PE . The inset shows the normalized plots at $T \geqslant 72$ K.

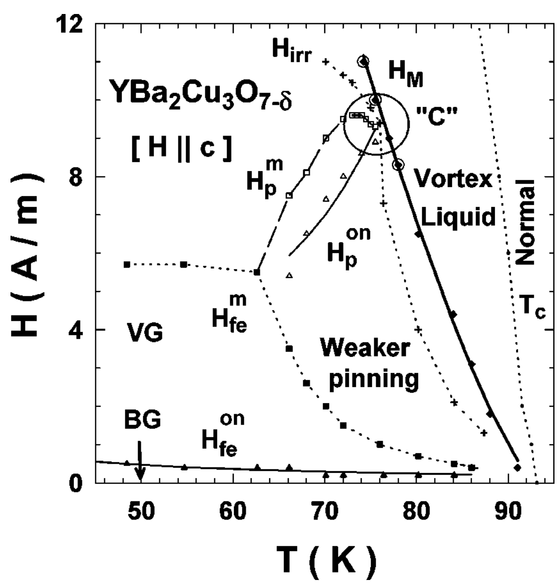

FIG. 4. A partial vortex phase diagram (for $H \| c$) in the given YBCO crystal. The noteworthy features are the existence of the first-order melting line $H_M(T)$ (as discerned from the jump in the equilibrium magnetization) well above the encircled point ''C,'' where the PE curve $[H_p^m(T)]$, the irreversibility line $[H_{\text{irr}}(T)]$, and the step change line $[H_m(T)]$ come closer. H_{fe}^{on} line is the boundary separating the Bragg glass (BG) from the vortex glass (VG). The thin and thick continuous lines are fits to the theoretical expressions (see text for a discussion). The lines passing through the respective H_{fe}^m, H_p^m, and H_{irr} data points are just guides to the eye.

Finally, the vortex phase diagram of YBCO in the low and intermediate fields is illustrated in Fig. 4. This phase diagram comprises a Bragg glass $(H < H_{fe}^{on})$, a vortex glass $(H_{fe}^{on} < H < H_{fe}^{m})$, a weaker pinned vortex solid $(H_{fe}^{m} < H < H_p^{on})$ and the vortex liquid $[H > H_{\text{step}}(\equiv H_M)]$. H_{fe}^{on} and H_p^{on} identify the onset fields of FE and PE, respectively. Note that the composite broad peak (FE) branches off around 65 K and as this happens, the upper curve (i.e., the PE curve) in Fig. 4 moves closer to the irreversibility line displaying an interesting reentrant characteristic. The thick continuous line marks the locus of the step change in the equilibrium magnetization, ΔM_{eq} . Note that the irreversibility line lies inside the step change curve, which has been designated as the melting curve H_M in Fig. 4. Near about 75 K, the section of irreversibility line comes very close to the reentrant portion of the PE curve (see circled region in Fig. 4). Such a behavior is also consistent with the earlier data of Nishizaki *et al.*[7] in an untwinned YBCO crystal. An important new feature in our present data is the branching of the so-called FE curve into an upper (PE) and a lower (FE) curve above 65 K. Another significant finding of our work is the extension of the melting (H_M) curve (i.e., the step change line) above the notional multicritical point ''C,'' where the PE curve, the step change curve and the irreversibility line appear to come very close. Since the step change is construed[7] as the signature of the first-order phase transition, the point ''C'' in Fig. 4 need not be a real tricritical point. This suggests that the real critical point and the possible second-order transition which is expected to occur beyond such a point, could be present well above the point ''C'' in Fig. 4.

Ertas and Nelson[20] and Kierfeld[21] have suggested that the temperature dependence of the disordering transition (onset of the PE) can be accounted by the relation $H_{\text{dis}}(T) = H_{\text{dis}}(0)(T^*/T)^{10/3}\exp[(2c/3)(T/T^*)^3]$ for $T > T^*$. T^* is defined as the single vortex depinning temperature and c is constant of the order unity and $H_{\text{dis}}(0)$ is the transition field for $T \leqslant T^*$. The thin continuous line in Fig. 4 indicates that the onset of PE data [i.e., $H_p^{on}(T)$] can be fitted fairly well to the above relation with $T^* = 54.72$ K and $H_{\text{dis}}(0) = 0.08$ T, which are comparable to the values of the parameters obtained in earlier studies.[7] For the case of the FLL melting line, an expression, $H_M = H_M(0)*(1-[T/T_c(0)])^n$, has been used. A fit to our data yields the thick continuous line, shown in Fig. 4. The values for the parameters $H_M(0)$ and n are found to be 94.6 T and $n = 1.34$, respectively. These values are in satisfactory agreement with those reported by others.[7] Further, we have used a relation, $H_{vg} = H_{vg}(0)T^{\beta}$ to fit the Bragg glass to the vortex glass transition line [filled triangle representing the $H_{fe}^{on}(T)$ data points in Fig. 4] and the continuous line is the resultant fit with $H_{vg}(0) = 11$ T and $\beta = -1.3$. These values compare reasonably with the values of 17.6 T and -1.09, respectively, as reported by Pissas *et al.*[22]

Recently, Kierfeld and Vinokur[16] have a proposed theory to account for the existence of a critical end point of the first-order melting line in the presence of point pins. They have assumed that the FLL melting is due to the dislocation mediated mechanism, in which each vortex state is characterized by its inherent dislocation density or, equivalently, by the dislocation spacing R_D. For the Bragg glass, $R_D=\infty$, and for the vortex liquid, $R_D=a_0$ (where a_0 is the lattice spacing), while $R_D=R_a$ for the vortex glass, where R_a is the positional correlation length. According to this theory, the coexistence point (where the topological transition meets the melting line) exists at a field, which is only 60% of the field, where the critical point (first-order to second-order transition) occurs in YBCO. However, our present studies cannot locate this critical point, since it is estimated to be about 16 T.

Before concluding it is useful to recall that Erb *et al.*[23] have surmised that clustering of oxygen vacancies amounting to inhomogeneous distribution of unit cells of the composition $YBa_2Cu_3O_{6.5}$ in the matrix of $YBa_2Cu_3O_7$ in a single crystal of $YBa_2Cu_3O_{7-\delta}$ ($\delta\neq 0$) could cause the broad fishtail effect anomaly. Pinning behavior of YBCO samples therefore could get governed by point pins, clustered vacancies and the twin boundaries. It is well known that the bulk $YBa_2Cu_3O_{6.5}$ becomes normal in the temperature range 60–70 K. Therefore, to extend the assertion of Erb *et al.*,[23] it may be speculated that the pinning characteristics, attributable to microscopic presence of $YBa_2Cu_3O_{6.5}$ unit cells in a YBCO specimen, undergo a change as the temperature varies through the 60 to 70 K range. Such a scenario may account for the splitting above 65 K of the composite broad anomaly in our crystal of YBCO.

In summary, we have reported the occurrence of the first-order transition well above the notional tricritical point, where the topological transition (represented by the PE) meets the FLL melting line. Usually, the presence of the twin boundaries in YBCO is considered to be detrimental to the observation of a first-order FLL melting transition. However, in our case, the presence of (few) twin boundaries appears to have facilitated the observation of the classical PE and the FE in the same isothermal scan in the intermediate field temperature range. The classical PE shows an interesting reentrant characteristic. The extension of the first-order melting line above the *notional coexistence point* could be construed as raising doubt about the validity of a ‘‘tricritical point,’’ and further studies are needed for quantitative comparison with recent works.[16]

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